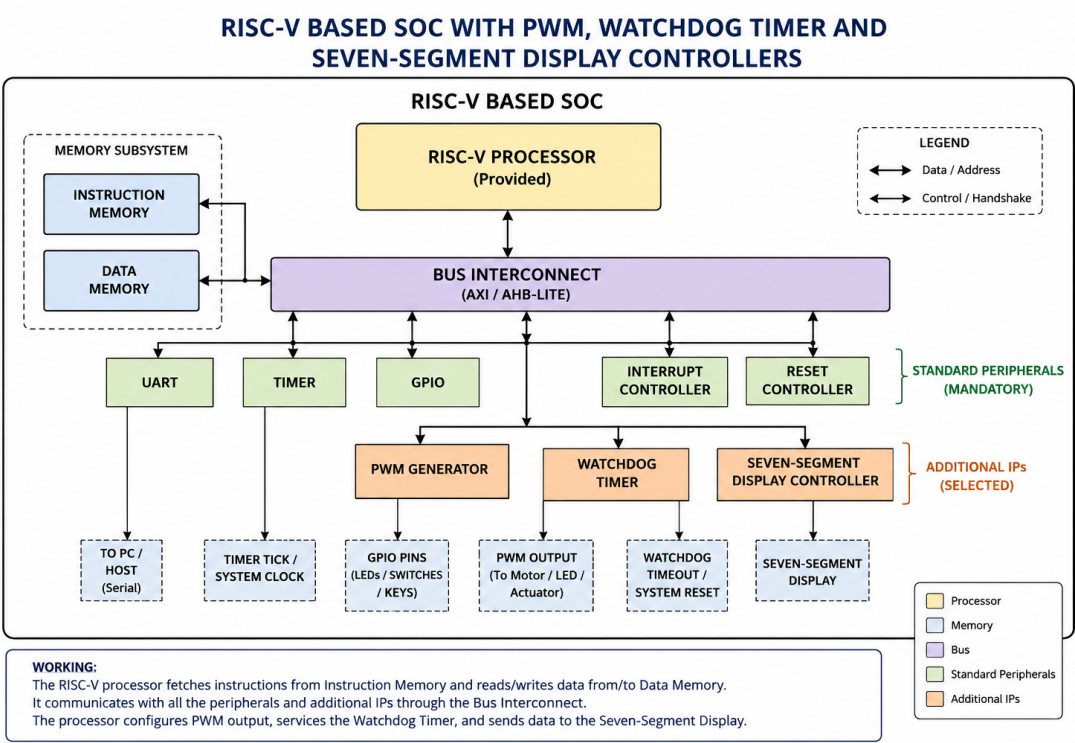


RISC-V BASED SOC WITH PWM, WATCHDOG TIMER AND SEVEN-SEGMENT DISPLAY CONTROLLERS

SoC Block Diagram



Top-Level SoC Boundary Signals

Signal	Direction w.r.t. SoC	Width	Purpose
clk	Input	1	Main system clock
rst_n	Input	1	System reset
uart_rx	Input	1	UART serial data received by SoC
uart_tx	Output	1	UART serial data transmitted by SoC
gpio_in	Input	N	External digital inputs
gpio_out	Output	N	External digital outputs
pwm_out	Output	1	PWM waveform generated by SoC
seg_out	Output	8	Seven-segment segment control
seg_digit_sel	Output	N	Selects digit in a multi-digit display
wdt_timeout	Output	1	Indicates watchdog timeout
wdt_reset_out	Output	1	Watchdog-generated reset request, if exposed